

Note 14.6 Tangent Planes and Differentials

Tangent Planes and Normal Lines

Definition

The tangent plane at a point is the plane through the point normal to the gradient vector.

The normal line of the surface at a point is the line through the point parallel to the gradient vector.

Expressions

Tangent Plane to $f(x, y, z) = c$ at $P_0(x_0, y_0, z_0)$:

$$\nabla f|_{P_0} \cdot \overrightarrow{P_0 P} = 0$$
$$\frac{\partial f}{\partial x}(x - x_0) + \frac{\partial f}{\partial y}(y - y_0) + \frac{\partial f}{\partial z}(z - z_0) = 0$$

Normal Line to $f(x, y, z) = c$ at $P_0(x_0, y_0, z_0)$:

$$\overrightarrow{P_0 P} = t \nabla f|_{P_0}$$
$$x = x_0 + \frac{\partial f}{\partial x}t, y = y_0 + \frac{\partial f}{\partial y}t, z = z_0 + \frac{\partial f}{\partial z}t$$

When encountering the surface $z = f(x, y)$, we just need to turn it to $F(x, y, z) = f(x, y) - z$.

Estimating Change in a Specific Direction

The function value increment equals to the product of directional derivative and distance increment.

$$df = (\nabla f|_{P_0} \cdot \mathbf{u}) ds$$

It can be obtained by the formula of directional derivative in dot product.

How to Linearize a Function of Two Variables

Definition

The linearization of a function of a function $f(x, y)$ at a point (x_0, y_0) where f is differentiable is

$$L(x, y) = f(x_0, y_0) + \frac{\partial f}{\partial x}(x - x_0) + \frac{\partial f}{\partial y}(y - y_0)$$

and $f(x, y) = L(x, y)$ is the standard linear approximation.

It can be analogy to the single-variable linearization. And it is obtained by the definition of differentiation.

And the plane $z = L(x, y)$ is tangent to the surface $z = f(x, y)$ at (x_0, y_0)

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$$\frac{\partial f}{\partial x}(x - x_0) + \frac{\partial f}{\partial y}(y - y_0) - (L(x, y) - f(x_0, y_0)) = 0$$

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Differentials

Definition

If we move from (x_0, y_0) to $(x_0 + dx, y_0 + dy)$ nearby, then

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy$$

in the linearization of f is called the total differential of f .

It can be obtained by the equations of linearization and replace by df, dx, dy .

Functions of More than Two Variables

Similar to upper content.

Exercises

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